

Topology General Exam

August 15, 2022

Instructions: This is a four hour exam. Your solutions should be legible and clearly organized, written in complete sentences in good mathematical style on your own paper. All work should be your own-no outside sources are permitted-using methods and results from the first year topology course topics. Each problem is worth the same number of points.

Problem 1

Let $\gamma : \mathbb{R} \rightarrow \mathbb{R}^n$ be a smooth curve. Prove that the set of real numbers

$$K = \{r > 0 \mid \text{the sphere of radius } r \text{ around } 0 \in \mathbb{R}^n \text{ is tangent to the image of } \gamma\}$$

has measure zero in $\mathbb{R}$.

Problem 2

(a) Let M and N be connected compact smooth manifolds of the same dimension (without boundary), and $f : M \rightarrow N$ a submersion. Prove that f is a covering map.

(b) Suppose M is a connected closed surface (2-manifold), and $f : M \rightarrow S^2$ a submersion. Prove that f must, in fact, be a diffeomorphism.

Problem 3

Consider the subset S of $\mathbb{R}^4$ defined by the two equations

$$\begin{aligned}x^2 + y^2 - z^2 - w^2 &= 1 \\xy + z + w &= 3\end{aligned}$$

Prove that S is a smooth manifold and find its dimension.

Problem 4

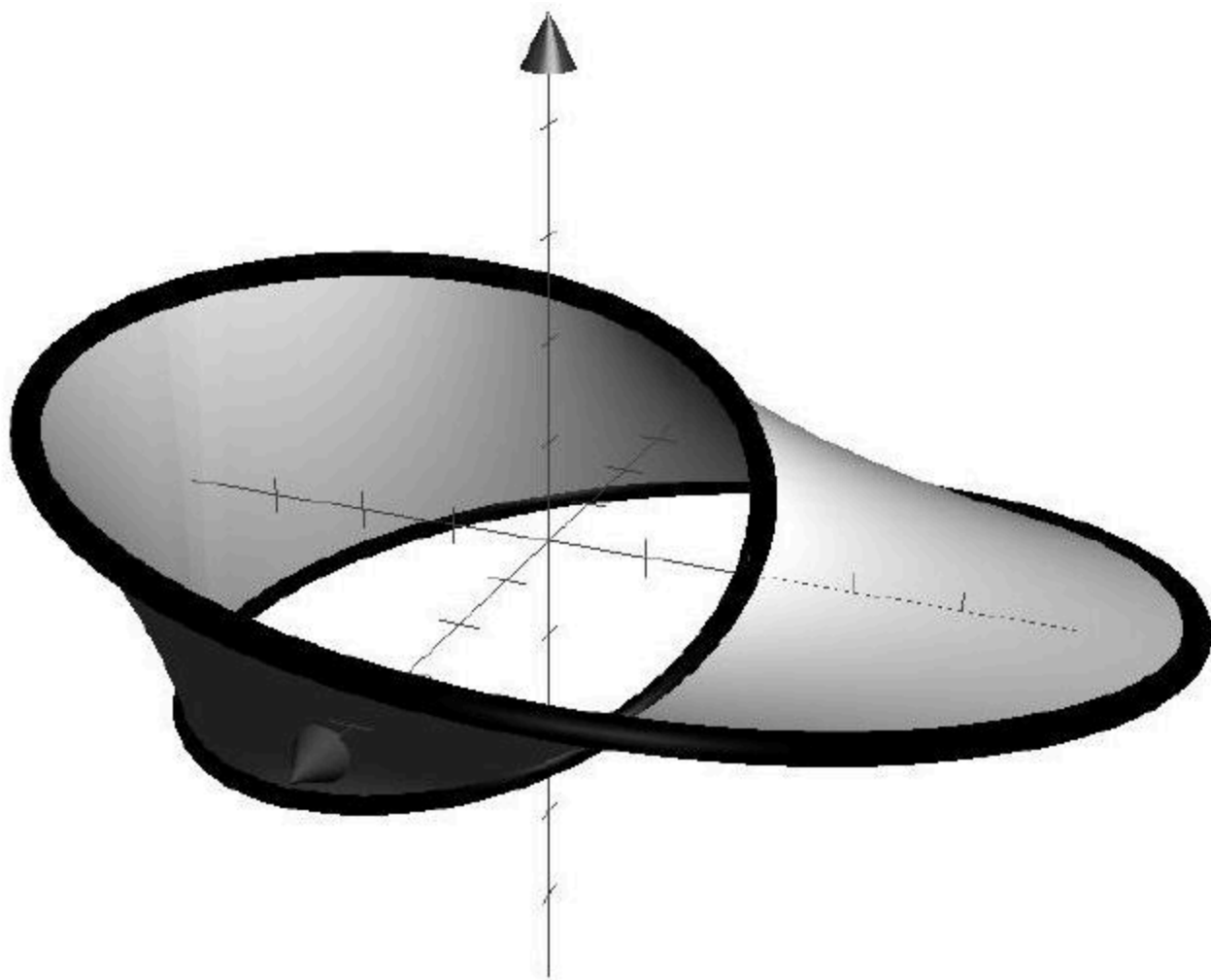
Consider the smooth, closed curve γ in $\mathbb{R}^3$ given in cylindrical coordinates $\underline{1}$ by

$$\gamma(t) = (r(t), \theta(t), z(t)), \quad \begin{aligned} r(t) &= 2 + \sin t \\ \theta(t) &= 2t \\ z(t) &= \cos t \end{aligned}$$

where $t \in [0, 2\pi]$. Visually, γ can be pictured as the boundary of a Möbius band $M \subset \mathbb{R}^3 - \{r = 0\}$ pictured below. Let α be the 1-form on $\mathbb{R}^3 - \{r = 0\}$ defined by $\alpha = d\theta$.

a) Find $\int_{\gamma} \alpha$.

b) Use your answer to (a) to prove that Möbius band M is not orientable, and in fact γ is not the boundary of any smooth, compact, orientable surface in $\mathbb{R}^3 - \{r = 0\}$.



The shaded Möbius band is a strip with one half-twist, embedded around the vertical axis without meeting that axis. Its single boundary curve γ is drawn in thick black; it runs along both edges of the strip before closing. The formula in the question specifies the parametrization and orientation of γ .

Problem 5

- (a) Show that if $f : S^4 \rightarrow S^4$ is continuous, then $f \circ f : S^4 \rightarrow S^4$ must have a fixed point.
 (b) By contrast, show that there is a continuous map $f : S^3 \rightarrow S^3$ such that $f \circ f$ has no fixed point.

Problem 6

Let X_n denote the n -skeleton of a CW complex X .

- a) Complete the definition: The cellular chain complex (C_*^{CW}, d_*) of X is defined by letting $C_n^{CW}(X) = H_n(X_n, X_{n-1})$ and then letting $d_n : C_{n+1}^{CW}(X) \rightarrow C_n^{CW}(X)$ be [you complete the definition].
 b) Using your definition from part (a), show that $d_{n-1} \circ d_n = 0$ for all $n \geq 1$.
 c) Prove that $C_n^{CW}(X)$ is isomorphic to a free abelian group with one generator for each n -cell of X .

Problem 7

Let $S^3 \xrightarrow{i_1} S^3 \vee S^3 \xrightarrow{i_2} S^3$ be the two inclusion maps into the wedge. Say that a map $f : S^3 \vee S^3 \rightarrow S^3$ has type (m, n) if the degree of $f \circ i_1$ is m and the degree of $f \circ i_2$ is n . Let $X_f = S^3 \cup_f (D^4 \vee D^4)$, i.e. X_f is the pushout

$$\begin{array}{ccc} S^3 \vee S^3 & \xrightarrow{f} & S^3 \\ \downarrow & & \downarrow \\ D^4 \vee D^4 & \longrightarrow & X_f. \end{array}$$

The square has $S^3 \vee S^3$ at the upper left, S^3 at the upper right, $D^4 \vee D^4$ at the lower left, and X with subscript f at the lower right. The top arrow is f from $S^3 \vee S^3$ to S^3 . The left arrow is the inclusion of the boundaries into $D^4 \vee D^4$. The remaining two arrows lead from S^3 and $D^4 \vee D^4$ into their pushout X with subscript f .

Compute the homology groups of X_f if f has type $(9, 6)$, describing the homology groups as direct sums of cyclic groups, as usual.

Problem 8

A group G is perfect if its commutator subgroup $[G, G]$ is all of G . The famous Poincaré 'sphere' is a 3-dimensional manifold M whose fundamental group is a perfect group of order 120. Show that any continuous function $M \rightarrow \mathbb{R}P^4$ must be null homotopic.

(You may assume that f is smooth if desired.)

1. Recall that r is the distance from the z -axis, and θ is the usual angle about the z -axis. ↩
2. The subgroup generated by elements of the form $ghg^{-1}h^{-1}$. ↩